

 Queen Mary University of London School of Business and Management	Coursework Assignment Instructions			Academic Year	
				2023/2024	
	Module Code: BUSM131			Level	7
	Module Title: Masterclass in Business Analytics				
Assessment Name: Essay			Weighting %	70	Group N
			Word Count	5000	

Module Organiser	Dr Guven Demirel	Date Submitted	
Internal Reviewer		Date Reviewed	
External Examiner		Date Approved	

Module Level Learning Outcomes to be assessed (Reference to Module Proposal Form)

No	Module Learning Outcome	Description
1	A1	Knowledge and understanding of a range of industry settings in which firms rely on business analytics, knowledge of the range of analytical tools being employed and understanding of their strengths and weaknesses.
2	A2	Knowledge and understanding of frameworks for data analysis in business contexts where large volumes of data are generated.
3	A3	Knowledge and understanding of the main intellectual frameworks in use in the fields of management and strategy and of key methodological issues involved in empirical research and data analysis in business contexts.
4	B1	Research an unfamiliar industry or firm context in order to establish the nature of an analytical problem
5	B2	Formulate an analytical strategy and to comment critically on analytical strategies adopted by others
7	C1	Competent to independently research a complex business contexts

AoL modules only: Programme Level Learning Outcomes contributed to (see programme rubric map)

No	Programme Learning Outcome	Description
1	LO 1.1	Make use of relevant and varied primary and/or secondary data (such as found in databases) in addressing the question at hand.
2	LO 1.2	Choose appropriate quantitative technique/s with clear justification.
3	LO 2.2	Demonstrate the ability to conduct a literature review, critically debate the literature, and evaluate contrasting viewpoints.
4	LO 2.3	Compare and contrast effectively differing methodologies, and justify the reasons why one has been selected for the task at hand.
5	LO 2.4	Apply creative problem solving and decision-making techniques in reaching a recommendation or conclusion.

6	LO 3.1	Cite sources of information and data consistently using an appropriate referencing style.
7	LO 3.2	Present findings as well-constructed argument that flows smoothly [applies to written work and oral presentations].

Assessment instructions for students (as per QMPlus 'Assessment Information' tab)

This essay is linked to the project that has the following components:

- the business context, business problem, decisions supported by analytics, and related business processes;
- one or more datasets that address the business problem;
- initial analysis of the datasets, including descriptive statistics and correlation;
- selection of tools to perform the analytical tasks;
- analysis of the datasets using Scikit-Learn tools following techniques suggested by Geron (2019); analysis of the dataset using Keras and Tensorflow;
- analysis of the results;
- application of the results to the business context, problem, processes and decisions;
- explanation of the application, results and tools to non-technical managers;
- discussion of the limitations of the project and suggested further work.

Each student works on one of the project topics and datasets:

1. Credit card applications
2. Credit card fraud detection
3. Hotel cancellations
4. Customer churn
5. Employee attrition
6. Bike sharing platform
7. Retail forecasting and planning
8. Used car listing price prediction
9. AirBnb price prediction
10. Hotel reviews analysis

Students have been given the ten project topics and data sets above from which they select one. Each student writes and submits a 5000-words essay (excluding code, graphs, tables, and appendices). The recommended number of words per section are added below.

Abstract (200 words)

- Brief summary of your project, including the business problem you study, the dataset you use, the machine learning models and techniques you use, the key results, and your solution to the problem.

Introduction (400 words)

- This section will provide a summary of the problem you are studying.
- A brief literature review on academic research on the topic (for instance, if your project is on customer churn, articles about predicting customer churn). Discuss at least three academic articles on the topic that use or focus on ML methods. Literature review articles are generally good resources.
- Summary of the structure of the report.

Business Problem (600 words)

- Explain the business context, the business problem, the decisions to be supported by analytics, and related business processes. Use information on company website, white papers, news articles, industry reports, case studies, etc. ("grey literature") to enrich your discussion about the business and explain clearly which business problem and processes your project is addressing.
- Mention some other organisations that face and solve similar problems using business analytics (for instance, if your project is on sports analytics mention some sports organisations investing into ML). Aim to provide three references to the "grey literature" for this purpose.

Data, EDA, and Methods (1500 words)

- Explain your dataset, including the features and target variables, datatypes, and missing values. If you use data from several sources, e.g. you scrape some data from web, explain your data collection and merging process.
- Provide descriptive statistics tables (univariate and bivariate) and statistical graphs to explore and present the data.
- Explain your pipeline of pre-processing, providing the details of how you tackle the missing data problem, feature engineering process (how you generate features from unstructured data or transforming existing variables), and variable scaling.
- Provide test-train-validation split and cross-validation details.
- Explain briefly which ML models you use for the analysis of the business problem you are studying. Use multiple ML models (aim to use at least three models). In this section, explain the method generally and provide the details and the results in the following section.
- Justify your methodological choices, explaining your reasoning in pre-processing and model selection.
- Explain which Python ML libraries you used.
- Note: Provide all Python codes in the Appendix and comment them clearly so that we can compare this with the corresponding analysis in the report. Cross-referencing in the main text is also recommended.

Analysis and Results (1500 words)

- Report the results from your analysis. Explain the details of how you specify each model (hyperparameters you use). Interpret the results you obtain by analysing relevant metrics and plots. Discuss what these mean for the specific problem you are analysing.
- Do not report each and every trial with a model. Show your important results and explain in the main text your choices, for instance how you chose the network architecture, activation functions, etc.
- Use tables as much as possible to compare results, for instance between different ML models or variants of a model family, e.g. different hyperparameter choices.
- Apply cross-validation to choose between initial models, then fine tune the hyperparameters of these models, also consider ensembling their predictions
- Use statistical graphs to communicate your results.
- Explain your reasoning in moving from one model to the other. For instance, you observe underfitting and then your next model aims to address this or your model fails to generalise nonlinear patterns, hence you use ensemble methods or deep learning.

Discussion and Conclusions (800 words)

- Summarise your technical results.
- Discuss and explain your results to the problem owner (manager). How does your proposed approach solve the problem? How much improvement can you bring, e.g. improved forecasting reduces costs by £2 million in comparison to traditional methods such as exponential smoothing.
- Consider which numbers, tables, and graphs best communicate your approach and results to your non-technical audience. You might want to develop some charts / diagrams for this section to explain your approach and results.
- What are the organisational changes required to implement your solution? Which business processes and decisions will be affected? How can you convince the parties affected?
- What are the limitations of your study and how can it be developed further? For instance, what further data would be desired and how can it be collected?

3. Assessment rubric with weighted criteria

10%	Abstract & Introduction	Does not clearly describe problem and/or its significance; no literature review is provided; no meaningful abstract	Superficial description of the business problem and/or its significance; summarizes the literature, but shows some confusion or inaccuracies; no clear connections between literature and the business problem; acceptable abstract	Clearly describes business problem and its significance; summarizes the literature but tends to simplify matters; the business problem is related to the studies but connections could be better explained; decent abstract	Clearly and persuasively describes the business problem and its significance; provides an excellent short literature review; clearly connects to the business problem that flows logically from the discussion; excellent abstract
		Poor business problem choice which cannot be addressed by ML methods; no meaningful discussion of the business context, problem, and processes; does not use secondary sources to frame the problem; does not relate to similar problems faced by other organisations.	Good business problem choice for which ML has a potential to help; superficial discussion of the business context, problem, and processes; poor use of secondary sources to frame the problem; does not clearly relate to similar problems faced by other organisations.	Very good business problem choice for which ML has a good potential to help; detailed and relevant discussion of the business context, problem, and processes; decent use of secondary sources to frame the problem; clearly relates to similar problems faced by other organisations.	Excellent business problem choice for which ML has a great potential to help; detailed discussion and elaboration of the business context, problem, and processes; excellent use of secondary sources to frame the problem; clearly and accurately relates to similar problems faced by other organisations.
		Poor data choice; No proper systematic pre-processing; Poor or no EDA; ML models chosen do not fit the problem; No discussion of methodological choices; Poor or no use of tables and graphs.	Acceptable data choice with some limitations; Pre-processing undertaken but either not clearly explained; Informative EDA; ML models are fine for the problem but there are better models; Some discussion of methodological choices but weaknesses in considerations; Acceptable use of tables and graphs (formatting can be improved and not sufficiently used).	Very good data choice, which is rich enough in terms of features and sample size but more features or larger dataset would be desirable; Good pre-processing undertaken and clearly explained; Informative EDA; ML models chosen fit the problem very well; Decent discussion of methodological choices; Good use of tables and graphs (formatting can be improved).	Excellent data choice which is rich enough in terms of features and sample size; Excellent use of further resources and pre-processing for feature engineering (if relevant); Robust and careful pre-processing undertaken; Informative and insightful EDA; ML models chosen fit the problem very well Convincing discussion of methodological choices; Excellent
15%	Business Problem				
25%	Data, EDA, and Methods				

					use of tables and graphs
25%	Analysis and Results	Poor choice and use of ML methods, techniques, and Python libraries; No proper discussion of the results; Not directly relevant to the business problem	Acceptable choice and use of ML methods, techniques, and Python libraries but clear limitations; Some but limited discussion of the results; No clear methodological discussions; No hyperparameter tuning; Some results relevant to the business problem; Acceptable use of tables and graphs	Very good choice and use of ML methods, techniques, and Python libraries; Provides a decent discussion on the results; Some decent methodological considerations but tends to simplify things; Some hyperparameter tuning; Good results that help solving the business problem; Good use of tables and graphs	Excellent choice and use of ML methods, techniques, and Python libraries; Elaborates on the results, analyses and presents them using proper metrics and graphs; Excellent methodological considerations in choosing between models; Very good use of hyperparameter tuning; Excellent results that satisfy the business goal; Excellent use of tables and graphs
10%	Discussion and Conclusions	Missing summary of study findings, limitations of study, and/or business problem relevance	Unclear summary of study findings, limitations of study, and/or connection to the business problem; not a clear communication strategy for the non-technical audience; implications do not flow logically from results.	Summarizes study findings; relates findings to the business problem; presents a solution to the business problem by ML methods, but the solution or the presentation can be improved; discusses the limitations of study but not clearly aware of limitations.	Clearly summarizes study findings; relates findings to the business problem; persuasively presents a solution to the business problem by ML methods, communicating the approach in a way understandable by problem owners; discusses the limitations of study and does not overstate claims.
15%	Structure and clarity of expression	Lacking structure; difficult to understand; poor referencing	Confusingly structured and argued; meaning unclear	Clearly structured paper; meaning generally clear, but some typos and grammatical errors	Clearly structured paper; persuasively written

Submission Information

Please observe the following style guide. Unless otherwise specified,

- All work must be typed and submitted in MS Word or Adobe PDF format
- Font size should be 12 point (unless otherwise specified)
- Font style should be Arial
- Lines should be double-spaced
- Leave margins for comment Insert page numbers

- Use a header containing your student ID number, the module code to which your work applies, and the date. And Please:
- Always spell-check and proof read your work before handing it in (once you have submitted your work you will not be permitted to retrieve it)
- Keep your own electronic back-up copy of your work and if possible save on two devices

Avoid plagiarism

- Submit your work on time

Guidelines and Late-Work Policy

1. Coursework submitted late (and there are no extenuating circumstances) will incur a late penalty. Five per
2. cent of the total marks available shall be deducted for every period of 24 hours, or part thereof, that an assignment is overdue there shall be a deduction of five per cent of the total marks available (i.e. five marks for an assessment marked out of 100). After seven calendar days (168 hours or more late) the mark shall be reduced to zero, and recorded as OFL (zero, fail, late).
3. Each module has word limits for coursework assignments; however, the decision about whether to impose a penalty mark for exceeding the word limit is made by each module organiser. You must check the module handbook and the assignment briefing documents to see whether the particular module organiser has adopted a penalty system. It is your responsibility to read the handbook and assignment briefing carefully. If no penalty is specified then the module organiser will take into account the word length under the standard marking conventions. For example, if you have exceeded the word limit then it might be that you have not been sufficiently succinct or focused in your assignment and therefore might be penalised for these weaknesses. Please note that word limits do NOT include references or appendices. However if excessive material is included on appendices then this too will be judged accordingly and you may be awarded a lower mark.
4. You should ensure that your submission is in either Microsoft Word or PDF format.
5. Failure to submit in either one of these formats will result in a mark of 0 being awarded for the particular assessment. It is therefore your responsibility to ensure that the file format is correct and it can be opened by the receiving party.
6. You should ensure that the correct piece of assignment is uploaded as the document downloaded on the due date by the module organiser will be marked regardless of content. You will not have another opportunity to submit the work again if you mistakenly uploaded the wrong document.
7. ALLOW YOURSELF PLENTY OF TIME TO SUBMIT YOUR COURSEWORK. DO NOT LEAVE IT UNTIL THE LAST MINUTE
8. Computer problems, such as computer viruses, failure to make a back-up copy or temporary internet access problems, will NOT be viewed as a valid reason for late submission.
9. Check that your assignment submission has been successful, and print a copy of the confirmation screen.
10. If you submit your assignment after the deadline, you will still be able to submit your coursework via QMplus however you will be penalised for late submission, the only exception to this is if you have an approved extension due to extenuating circumstances.

Submission Deadline

Date	
Time	3pm
QMplus Link/Page	

Help and Support

- Module Organiser
- Disability and Dyslexia Service (DDS)

- Quantitative Skills Tutor
- Academic Writing Skills Tutor